Lab Assignment - 8

Protein Sequence Analysis - Motif Detection

BT3040 - Bioinformatics

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Question 1

The Python code to determine the hydrophobicity profiles, and identify the α-helices and β-strands for the sequences in the file `Q1.fasta` is given below:

```
# Question 1
def sec_str_identification(seq, seq_id):
  seq_scores = []

  for AA in seq:
    seq_scores.append(hydrophobicity[AA])
  seq_scores = np.array(seq_scores)
  mean_score = np.mean(seq_scores)
  characterization = np.where(seq_scores > mean_score, 1, -1)
  alpha_helix, beta_strand = set(), set()

  # Identifying alpha-helices
  n = 0
  while n < (len(characterization)-3):
    if (characterization[n] == characterization[n+1]) and
    (characterization[n+2] == characterization[n+3]) and
    ((characterization[n] * characterization[n+2]) < 0):

      alpha_helix.update([n, n+1, n+2, n+3])
      n = n+2
    else:
      n = n+1

  # Identifying beta-strands
  n = 0
  while n < (len(characterization)-3):
    if (characterization[n] == characterization[n+2]) and
    (characterization[n+1] == characterization[n+3]) and
    ((characterization[n] * characterization[n+1]) < 0):

      beta_strand.update([n, n+1, n+2, n+3])
      n = n+2
    else:
      n = n+1

  # Plotting the graph
  x = np.arange(1, len(seq_scores) + 1)

  fig, ax = plt.subplots(figsize=(10, 4), dpi=200)

  ax.plot(x, seq_scores, c="black", marker="o", markersize= 2.5,
  label="Hydrophobicity")
  ax.axhline(mean_score, c="black", ls="--", lw=0.75, label="Mean")

  for pos in set(alpha_helix):
      ax.axvspan(pos + 0.5, pos + 1.5, alpha=0.5, color="blue")
```

```
  for pos in set(beta_strand):
      ax.axvspan(pos + 0.5, pos + 1.5, alpha=0.5, color="red")

  helix_patch = mpatches.Patch(color='blue', alpha=0.5, label='alpha-helix')
  sheet_patch = mpatches.Patch(color='red', alpha=0.5, label='beta-strand')

  ax.set_title(f"Hydrophobicity profile of {seq_id}", fontsize=10)
  ax.set_xlabel("Residue number", fontsize=10)
  ax.set_ylabel("Hydrophobicity value", fontsize=10)

  ax.legend(handles=[helix_patch, sheet_patch],
            loc='upper left',
            bbox_to_anchor=(1.02, 1),
            borderaxespad=0.)

  plt.tight_layout(rect=[0, 0, 0.85, 1])

  plt.show()
  plt.close(fig)

# Executing the function:
Sequences = {}

for record in SeqIO.parse("/content/drive/MyDrive/BT3040-Assignments/
Lab Assignment - 8/Q1.fasta", "fasta"):
  Sequences[record.id] =  str(record.seq)

for sequence_id in Sequences.keys():
  sequence = Sequences[sequence_id]
  sec_str_identification(sequence, sequence_id)
```

The output of the program is shown below:

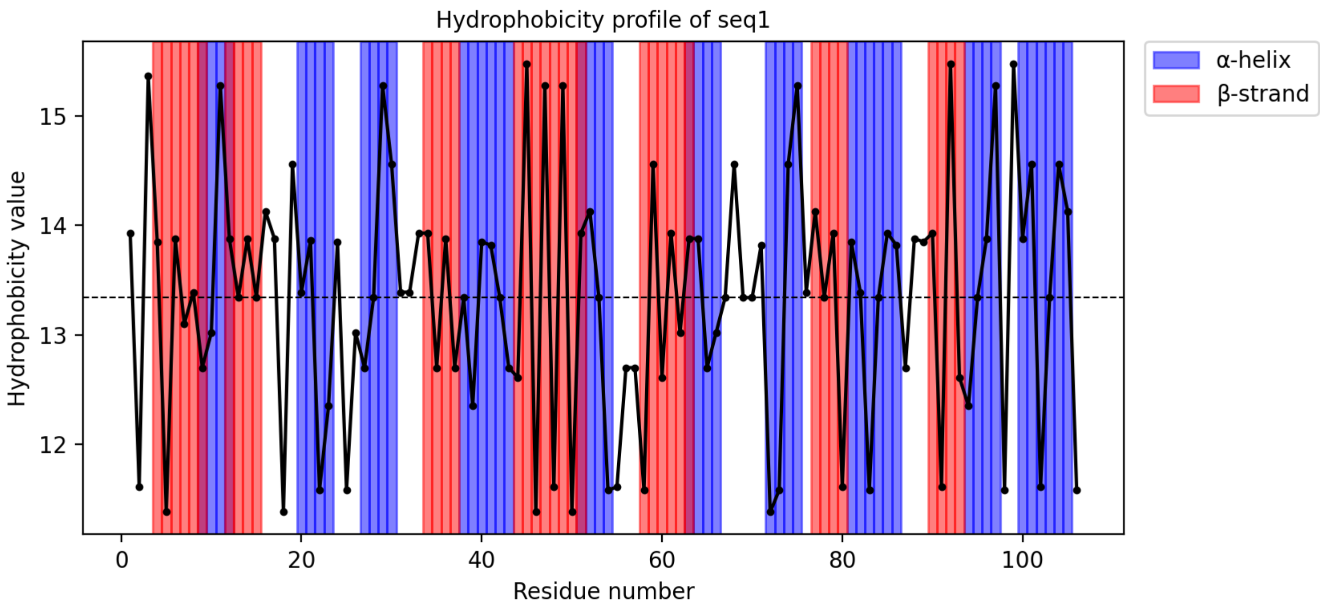

Figure 1.1: Hydrophobicity profile of sequence 1

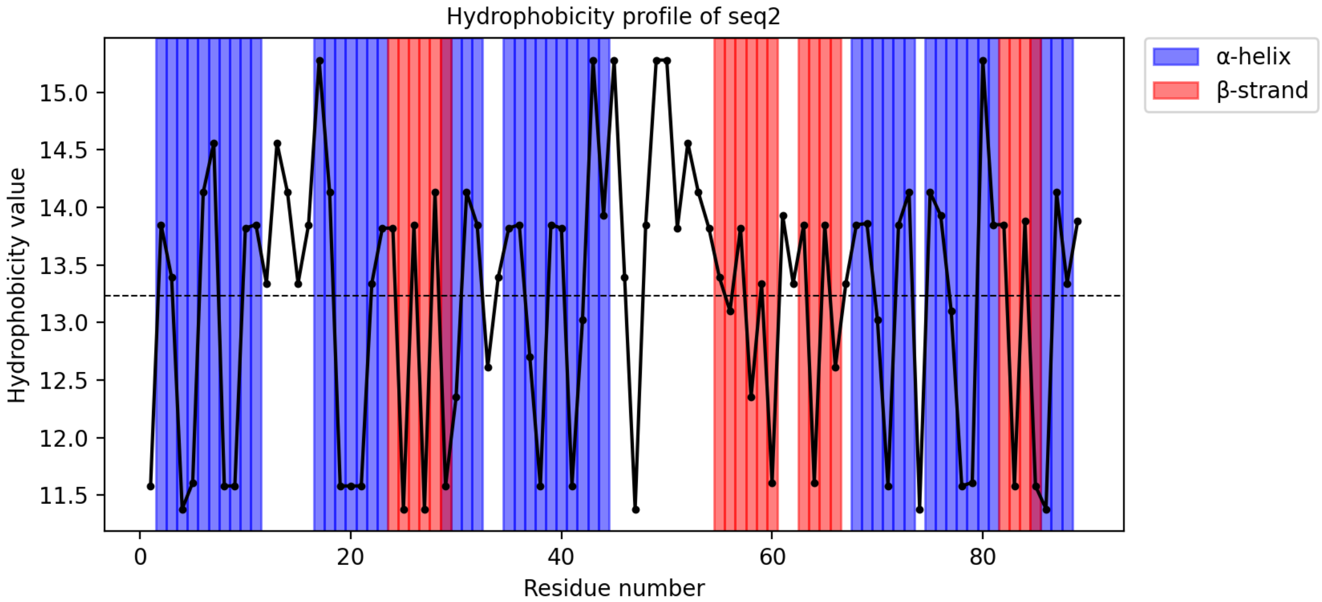

Figure 1.2: Hydrophobicity profile of sequence 2

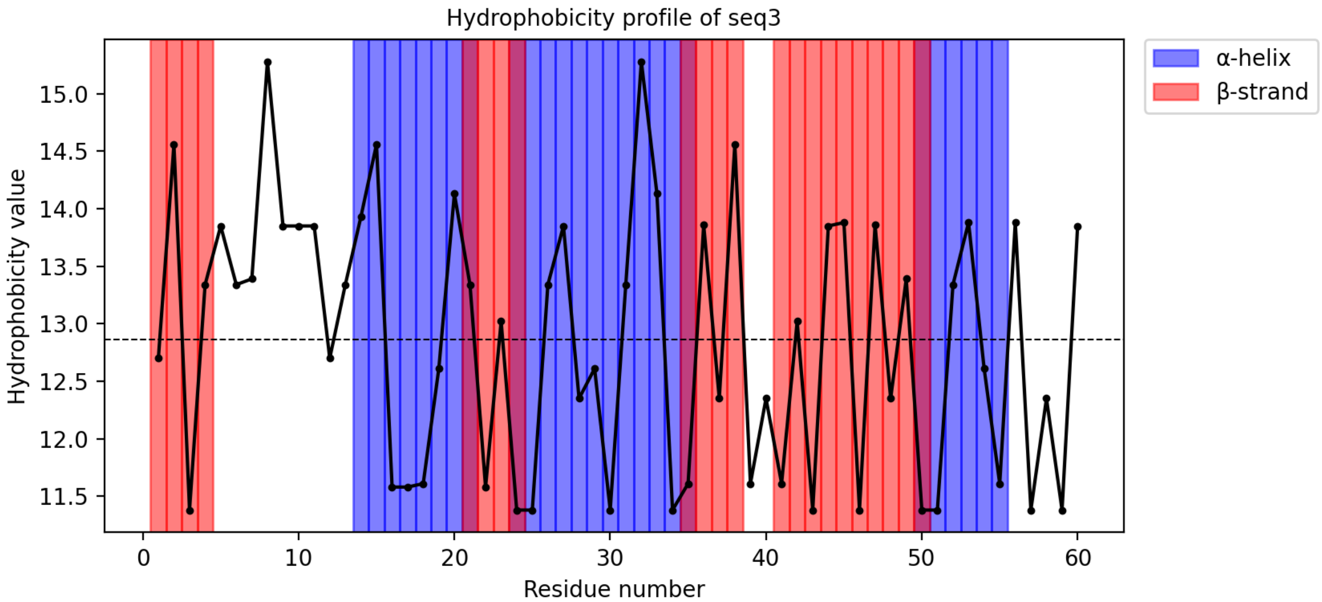

Figure 1.3: Hydrophobicity profile of sequence 3

Question 2

The Python code to calculate the amphipathic indices for the helices (of length 8) and the strands (of length 6) found in Q1 is given below:

```
# Question 2
def helix_amphipaticity_scoring(frames, seq):
  output = {}
  for frame in frames:
    subseq = (seq[frame[0]:frame[1]+1])

    a1 = (hydrophobicity[subseq[0]]+hydrophobicity[subseq[4]])/2
```

```
    a2 = (hydrophobicity[subseq[1]]+hydrophobicity[subseq[5]])/2
    a3 = (hydrophobicity[subseq[2]]+hydrophobicity[subseq[6]])/2
    a4 = (hydrophobicity[subseq[3]]+hydrophobicity[subseq[7]])/2

    A1 = round(abs((a1 + a2) - (a3 + a4)),3)
    A2 = round(abs((a1 + a4) - (a2 + a3)),3)

    A_alpha = max(A1, A2)
    output[subseq] = A_alpha

  return(output)
#--------------------------------------------------------------------------------
def strand_amphipaticity_scoring(frames, seq):
  output = {}
  for frame in frames:
    subseq = (seq[frame[0]:frame[1]+1])

    b1 = (hydrophobicity[subseq[0]]+hydrophobicity[subseq[2]]+
        hydrophobicity[subseq[4]])/3

    b2 = (hydrophobicity[subseq[1]]+hydrophobicity[subseq[3]]+
        hydrophobicity[subseq[5]])/3

    B = round(abs(b1-b2),3)
    output[subseq] = B

  return(output)
#--------------------------------------------------------------------------------
# Getting the frames for amphipaticity score calculations
def amphipathicity(seq_id, seq):
  seq_scores = []

  for AA in seq:
    seq_scores.append(hydrophobicity[AA])
  seq_scores = np.array(seq_scores)
  mean_score = np.mean(seq_scores)
  characterization = np.where(seq_scores > mean_score, 1, -1)

  helix_chains, strand_chains = [], []

  # Identifying alpha-helices of length 8
  n = 0
  while n < (len(characterization)-7):
    if ((characterization[n] == characterization[n+1]) and
    (characterization[n] == characterization[n+4]) and
    (characterization[n] == characterization[n+5]) and
    (characterization[n+2] == characterization[n+3]) and
    (characterization[n+2] == characterization[n+6]) and
    (characterization[n+2] == characterization[n+7]) and
    (characterization[n] * characterization[n+2] < 0)):

      helix_chains.append((n,n+7))
```

```
    n += 1

  # Identifying beta-strands of length 6
  n = 0
  while n < (len(characterization)-5):
    if ((characterization[n] == characterization[n+2]) and
    (characterization[n] == characterization[n+4]) and
    (characterization[n+1] == characterization[n+3]) and
    (characterization[n+1] == characterization[n+5]) and
    (characterization[n] * characterization[n+1] < 0)):

      strand_chains.append((n,n+5))
    n += 1

  # For alpha-helices
  if helix_chains == []:
    print(f"No α-helix of 8 or more residues are present in {seq_id}")
  else:
    Helix_Frames = []
    for block in helix_chains:
      lower, upper = block
      n = lower
      while n <= upper-7:
        Helix_Frames.append((n,n+7))
        n+=2

    helix_amphipathicity_score = helix_amphipaticity_scoring(Helix_Frames,seq)
    for subseq in helix_amphipathicity_score.keys():
      print(f"The α-helix segment {subseq} has an amphipaticity
      index of: {helix_amphipathicity_score[subseq]}")

  # For beta-strands
  if strand_chains == []:
    print(f"No β-strands of 6 or more residues are present in {seq_id}")
  else:
    Strand_Frames = []
    for block in strand_chains:
      lower, upper = block
      n = lower
      while n <= upper-5:
        Strand_Frames.append((n,n+5))
        n+=1

    strand_amphipathicity_score = strand_amphipaticity_scoring(Strand_Frames,seq)
    for subseq in strand_amphipathicity_score.keys():
      print(f"The β-strands segment {subseq} has an amphipaticity
      index of: {strand_amphipathicity_score[subseq]}")

# Executing the function:
for seq_id in Sequences.keys():
  print(seq_id)
  amphipathicity(seq_id, Sequences[seq_id])
```

The output of the above code is given below:

```
seq1
No α-helix of 8 or more residues are present in seq1
The β-strands segment AEYRST has an amphipaticity index of: 1.313
The β-strands segment QWEIDI has an amphipaticity index of: 3.48
The β-strands segment WEIDIE has an amphipaticity index of: 3.89
The β-strands segment EIDIEF has an amphipaticity index of: 3.373
The β-strands segment KVQFNY has an amphipaticity index of: 1.72
================================================================================
seq2
The α-helix segment ASEDLVKK has an amphipaticity index of: 4.89
The α-helix segment EDLVKKHA has an amphipaticity index of: 5.105
The α-helix segment HATKAHKN has an amphipaticity index of: 3.23
The α-helix segment TKAHKNIF has an amphipaticity index of: 4.0
The β-strands segment HEAELK has an amphipaticity index of: 2.487
The β-strands segment SRHPGD has an amphipaticity index of: 1.163
The β-strands segment RHPGDF has an amphipaticity index of: 1.343
================================================================================
seq3
No α-helix of 8 or more residues are present in seq3
The β-strands segment YEMPSE has an amphipaticity index of: 2.007
================================================================================
```

Question 3

The hydrophobicity profile of the sequence from `Q2.fasta` was smoothed using sliding window sizes of 9 and 19, respectively. The code used to perform this is given below:

```
# Question 3
def transmembrane_domains(seq, win_len):
  seq_scores = []

  for AA in seq:
    seq_scores.append(hydrophobicity[AA])
  seq_scores = np.array(seq_scores)

  # Symmetric half-window
  half_win = win_len//2
  profile = []

  for n in range(len(seq_scores)):
      start = max(0, n - half_win)
      end = min(len(seq_scores), n + half_win + 1)
      window = seq_scores[start:end]
      profile.append(np.mean(window))

  # Getting the smoothed profile
  profile = np.array(profile)
  threshold = np.mean(profile)
  characterization = np.where(profile > threshold, 1, -1)
  transmembrane_regions = set()
```

```
  for n in range(len(characterization) - 10 + 1):
    if np.all(characterization[n:n+10] == 1):
        transmembrane_regions.update(range(n, n+10))

  # For the plot
  positions = np.arange(1, len(profile) + 1)
  fig, ax = plt.subplots(figsize=(10, 4), dpi=200)

  ax.plot(positions, profile, c="black", marker="o", markersize=1.5,
        label="Hydrophobicity")

  ax.axhline(threshold, c="black", ls="--", lw= 1, label="Mean")

  # Highlight TM regions
  for pos in set(transmembrane_regions):
      ax.axvspan(pos + 0.5, pos + 1.5, alpha=0.5, color="gold")

  # Legend
  tm_patch = mpatches.Patch(color='gold', alpha=0.5, label='Transmembrane')

  # Labels and title
  ax.set_xlabel("Residue position", fontsize=10)
  ax.set_ylabel("Hydrophobicity", fontsize=10)
  ax.set_title("Hydrophobicity Profile", fontsize=10)

  # Legend positioning
  ax.legend(handles=[tm_patch],
          loc='upper left',
          bbox_to_anchor=(1.02, 1),
          borderaxespad=0.)

  plt.tight_layout(rect=[0, 0, 0.85, 1])
  plt.show()
  plt.close(fig)

# Reading the fasta file!
Sequences = {}

for record in SeqIO.parse("/content/drive/MyDrive/BT3040-Assignments/
Lab Assignment - 8/Q2.fasta", "fasta"):
  Sequences[record.id] =  str(record.seq)

# Executing the function:
for seq_id in Sequences.keys():
  print(seq_id)
  print(Sequences[seq_id])
  transmembrane_domains(Sequences[seq_id], 9)
  transmembrane_domains(Sequences[seq_id], 19)
```

In the above code we classify regions with 10 consecutive residue that have a hydrophobicity index greater than the average hydrophobicity of the sequence to classify the trans-membrane domains. The plots obtained from the above code are shown below.

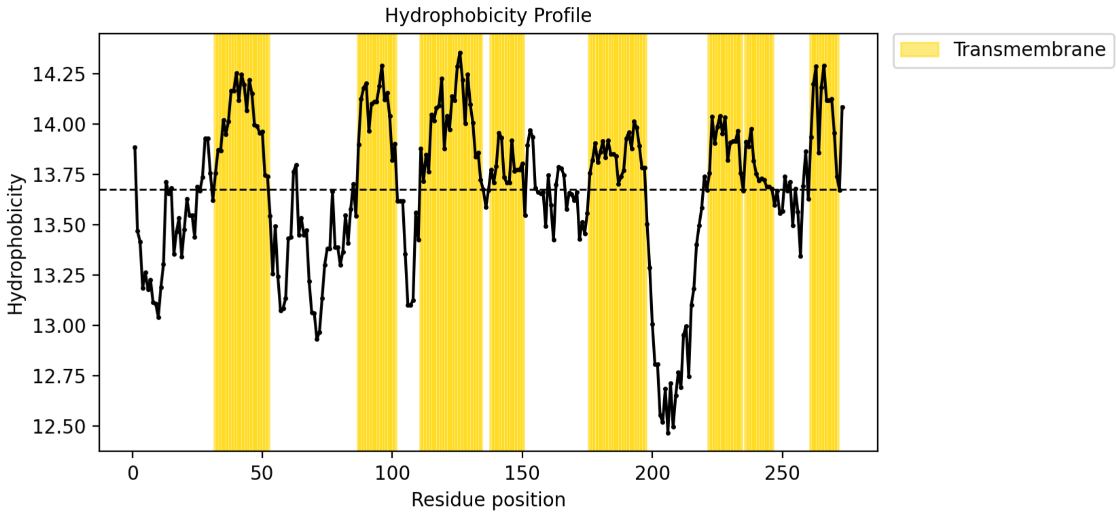

Figure 3.1: Hydrophobicity Profile - smoothing window size 9

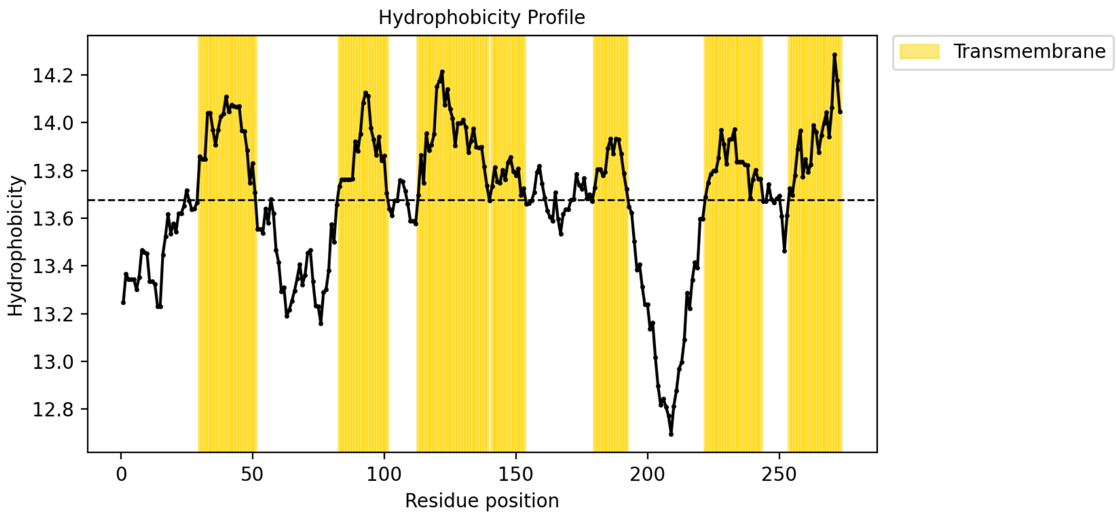

Figure 3.2: Hydrophobicity Profile - smoothing window size 19

Question 4

The ScanProsite tool was used to search for the given patterns in UniProtKB database.

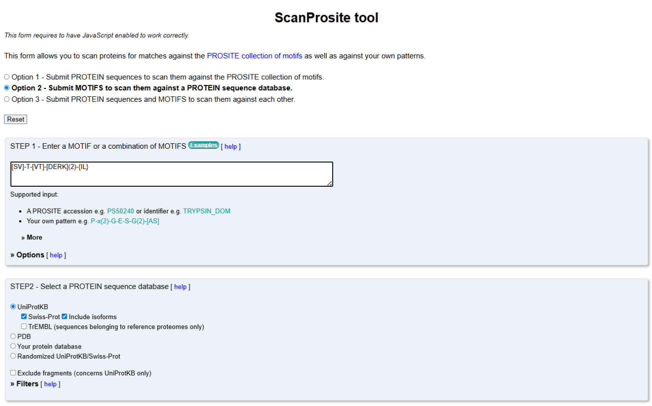

Figure 4.1: Pattern-1

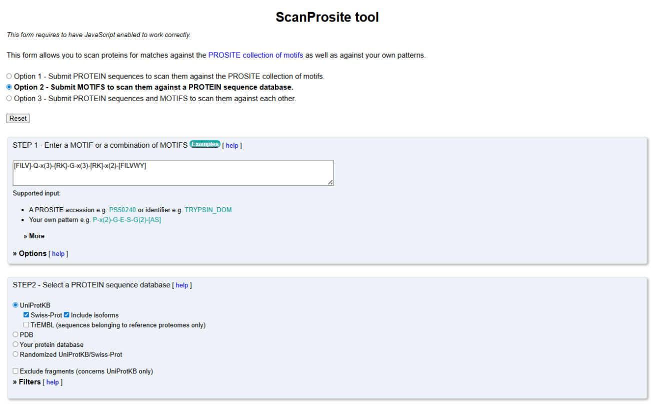

Figure 4.2: Pattern-2

The results of the ScanProsite tool is shown below:

[ScanProsite] Results for job: Pattern-1_Assignment-8_Question-4

Expasy <expasy@expasy.org>
To: be23b016@smail.iitm.ac.in

[Mon Mar 23 15:02:37 2026] - Submission
[Mon Mar 23 20:40:44 2026] - Results:

WARNING: results output is bigger than 4MB!
The output was truncated to 4MB.
You may try to change your scan parameters to get less results.

include splice variants (UniProtKB/Swiss-Prot)
Output format: Text

Hits for USER001 "[SV]-T-[VT]-[DERK](2)-{IL}" on UniProtKB/Swiss-Prot sequences
UniProtKB/Swiss-Prot (Release 2026_01 of 28-Jan-26) contains 574,627 entries.

Found: 11115 hits in 10688 sequences<hr>

Figure 4.1: Pattern-1, Results

[ScanProsite] Results for job: Pattern-2_Assignment-8_Question-4

Expasy <expasy@expasy.org>
To: be23b016@smail.iitm.ac.in

[Mon Mar 23 15:14:12 2026] - Submission
[Mon Mar 23 20:41:41 2026] - Results:

include splice variants (UniProtKB/Swiss-Prot)
Output format: Text

Hits for USER001 "[FILV]-Q-x(3)-{RK}-G-x(3)-[RK]-x(2)-[FILVWY]" on UniProtKB/Swiss-Prot sequences
UniProtKB/Swiss-Prot (Release 2026_01 of 28-Jan-26) contains 574,627 entries.

Found: 3882 hits in 3817 sequences<hr>

Figure 4.2: Pattern-2, Results

For pattern-1 the software found: 11,115 hits in 10,688 sequences.
For pattern-2 the software found: 3,882 hits in 3,817 sequences.

Question 5

The program to identify the patterns in `Q4.fasta`

```
pattern1 = re.compile(r'[SV]T[VT][DERK]{2}[^IL]')
pattern2 = re.compile(r'[FILV]Q.{3}[^RK]G.{3}[RK].{2}[FILVWY]')

for header in Sequences.keys():
  seq = Sequences[header]
  matches_1 = list(re.finditer(pattern1, seq))
  matches_2 = list(re.finditer(pattern2, seq))

  if matches_1 or matches_2:
    print(header)
    for m in matches_1:
      print(f"Pattern 1 | Position: {m.start()+1} | {m.group()}")
    for m in matches_2:
      print(f"Pattern 2 | Position: {m.start()+1} | {m.group()}")
    print("="*50)
```

The output of the above code is given below:

```
4A0C_2|Chains C,E|CULLIN-4B|HOMO SAPIENS (9606)
Pattern 1 | Position: 665 | STTERV
==================================================
4A0K_1|Chain A|CULLIN-4A|HOMO SAPIENS (9606)
Pattern 1 | Position: 666 | STTERV
==================================================
4FXG_2|Chains B,E|Complement C4-A alpha chain|Homo sapiens (9606)
Pattern 2 | Position: 252 | LQIEKEGAIHREEL
==================================================
4FXK_2|Chain B|Complement C4-A Alpha chain|Homo sapiens (9606)
Pattern 2 | Position: 252 | LQIEKEGAIHREEL
==================================================
```

```
4XAM_2|Chains C,E|Complement C4-A|Homo sapiens (9606)
Pattern 2 | Position: 175 | LQIEKEGAIHREEL
==================================================
5F0J_3|Chain C|Sorting nexin-3|Homo sapiens (9606)
Pattern 1 | Position: 70 | STVRRR
==================================================
5F0L_3|Chain C|Sorting nexin-3|Homo sapiens (9606)
Pattern 1 | Position: 70 | STVRRR
==================================================
5F0M_3|Chain C|Sorting nexin-3|Homo sapiens (9606)
Pattern 1 | Position: 70 | STVRRR
==================================================
5F0P_3|Chain C|Sorting nexin-3|Homo sapiens (9606)
Pattern 1 | Position: 70 | STVRRR
==================================================
5JPM_2|Chains B,E|Complement C4-A|Homo sapiens (9606)
Pattern 2 | Position: 252 | LQIEKEGAIHREEL
==================================================
5JPN_2|Chain B|Complement C4-A|Homo sapiens (9606)
Pattern 2 | Position: 252 | LQIEKEGAIHREEL
==================================================
5JTW_2|Chains B,E|Complement C4-A|Homo sapiens (9606)
Pattern 2 | Position: 175 | LQIEKEGAIHREEL
==================================================
5N69_1|Chains A,B|Myosin-7|Bos taurus (9913)
Pattern 1 | Position: 69 | VTVKED
==================================================
5N6A_1|Chain A|Myosin-7|Bos taurus (9913)
Pattern 1 | Position: 69 | VTVKED
==================================================
5TBY_1|Chains A,B|Myosin-7|Homo sapiens (9606)
Pattern 1 | Position: 69 | VTVKED
==================================================
6FSA_1|Chains A,B|Myosin-7|Bos taurus (9913)
Pattern 1 | Position: 69 | VTVKED
==================================================
6X5Z_3|Chains D,G,J|Myosin-7|Bos taurus (9913)
Pattern 1 | Position: 69 | VTVKED
==================================================
6YSQ_1|Chains A,B|Complement C4-B,Complement C4-B|Homo sapiens (9606)
Pattern 2 | Position: 854 | LQIEKEGAIHREEL
==================================================
7JH7_2|Chains F,G,H|Myosin-7|Sus scrofa (9823)
Pattern 1 | Position: 69 | VTVKED
==================================================
1A8J_1|Chains H,L|IMMUNOGLOBULIN LAMBDA LIGHT CHAIN DIMER (MCG)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1ADQ_2|Chain L|IGM-LAMBDA RF-AN FAB (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 201 | STVEKT
==================================================
1AIV_1|Chain A|OVOTRANSFERRIN|Gallus gallus (9031)
Pattern 1 | Position: 543 | STVEEN
==================================================
1AQK_1|Chain L|FAB B7-15A2|Homo sapiens (9606)
```

Pattern 1 | Position: 204 | STVEKT
==
1BJM_1|Chains A,B|LOC - LAMBDA 1 TYPE LIGHT-CHAIN DIMER|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==
1DCL_1|Chains A,B|MCG|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==
1DPU_1|Chain A|REPLICATION PROTEIN A (RPA32) C-TERMINAL DOMAIN|Homo sapiens (9606)
Pattern 1 | Position: 86 | STVDDD
==
1E8I_1|Chains A,B|EARLY ACTIVATION ANTIGEN CD69|HOMO SAPIENS (9606)
Pattern 1 | Position: 19 | STVKRS
==
1FM2_2|Chain B|GLUTARYL 7-AMINOCEPHALOSPORANIC ACID ACYLASE|Brevundimonas diminuta (293)
Pattern 1 | Position: 112 | STVDKP
==
1FM5_1|Chain A|EARLY ACTIVATION ANTIGEN CD69|Homo sapiens (9606)
Pattern 1 | Position: 40 | STVKRS
==
1FNT_15|Chains c,d,e,f,g,h,i,j,k,l,m,n,o,p|PROTEASOME ACTIVATOR PROTEIN PA26|Trypanosoma brucei (5691)
Pattern 1 | Position: 153 | STVEDK
==
1FO9_1|Chain A|ALPHA-1,3-MANNOSYL-GLYCOPROTEIN BETA-1,2-N-ACETYLGLUCOSAMINYLTRANSFERASE|Oryctolagus cuniculus (9986)
Pattern 1 | Position: 19 | STVRRC
==
1FOA_1|Chain A|ALPHA-1,3-MANNOSYL-GLYCOPROTEIN BETA-1,2-N-ACETYLGLUCOSAMINYLTRANSFERASE|Oryctolagus cuniculus (9986)
Pattern 1 | Position: 19 | STVRRC
==
1GPJ_1|Chain A|Glutamyl-tRNA reductase|Methanopyrus kandleri (2320)
Pattern 1 | Position: 316 | STVEEE
==
1IQ7_1|Chain A|Ovotransferrin|Gallus gallus (9031)
Pattern 1 | Position: 202 | STVEEN
==
1JT1_1|Chain A|FEZ-1, class B3 metallo-beta-lactamase|Fluoribacter gormanii (464)
Pattern 1 | Position: 122 | STVDKV
==
1JVK_1|Chains A,B|IMMUNOGLOBULIN LAMBDA LIGHT CHAIN|Homo sapiens (9606)
Pattern 1 | Position: 205 | STVEKT
==
1JVZ_2|Chain B|cephalosporin acylase beta chain|Brevundimonas diminuta (293)
Pattern 1 | Position: 112 | STVDKP
==
1K07_1|Chains A,B|FEZ-1 beta-lactamase|Fluoribacter gormanii (464)
Pattern 1 | Position: 122 | STVDKV
==
1KEH_1|Chain A|precursor of cephalosporin acylase|Brevundimonas diminuta (293)
Pattern 1 | Position: 281 | STVDKP
==
1KVD_2|Chains B,D|SMK TOXIN|Pichia farinosa (4920)

```
Pattern 1 | Position: 46 | STVKDG
==================================================
1KVE_2|Chains B,D|SMK TOXIN|Pichia farinosa (4920)
Pattern 1 | Position: 46 | STVKDG
==================================================
1L9Y_1|Chains A,B|FEZ-1 b-lactamase|Fluoribacter gormanii (464)
Pattern 1 | Position: 122 | STVDKV
==================================================
1LGV_1|Chains A,B|IMMUNOGLOBULIN LAMBDA LIGHT CHAIN|Homo sapiens (9606)
Pattern 1 | Position: 205 | STVEKT
==================================================
1LHZ_1|Chains A,B|IMMUNOGLOBULIN LAMBDA LIGHT CHAIN|Homo sapiens (9606)
Pattern 1 | Position: 205 | STVEKT
==================================================
1LIL_1|Chains A,B|LAMBDA III BENCE JONES PROTEIN CLE|Homo sapiens (9606)
Pattern 1 | Position: 200 | STVEKT
==================================================
1MCB_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCC_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCE_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCF_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCH_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCJ_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCL_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCO_1|Chain L|IGG1 MCG INTACT ANTIBODY (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCR_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCS_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1NL0_1|Chain L|anti-factor IX antibody, 10C12, chain L|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1OVT_1|Chain A|OVOTRANSFERRIN|Gallus gallus (9031)
Pattern 1 | Position: 543 | STVEEN
==================================================
1Q1J_1|Chains L,M|Fab 447-52D, light chain|Homo sapiens (9606)
Pattern 1 | Position: 205 | STVEKT
```

```
=====================================================
1R5M_1|Chain A|SIR4-interacting protein SIF2|Saccharomyces cerevisiae (4932)
Pattern 1 | Position: 12 | STVERE
=====================================================
1RP1_1|Chain A|PANCREATIC LIPASE RELATED PROTEIN 1|Canis lupus familiaris (9615)
Pattern 1 | Position: 437 | STVRED
=====================================================
1RYX_1|Chain A|Ovotransferrin|Gallus gallus (9031)
Pattern 1 | Position: 543 | STVEEN
=====================================================
1RZF_1|Chain L|Fab E51 light chain|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
=====================================================
1S5J_1|Chain A|DNA polymerase I|Sulfolobus solfataricus (2287)
Pattern 1 | Position: 604 | STVKKA
=====================================================
1UYP_1|Chains A,B,C,D,E,F|BETA-FRUCTOSIDASE|THERMOTOGA MARITIMA (243274)
Pattern 1 | Position: 371 | STVEDE
=====================================================
1VQT_1|Chain A|Orotidine 5'-phosphate decarboxylase|Thermotoga maritima (2336)
Pattern 1 | Position: 72 | STVERS
=====================================================
1W72_5|Chains L,M|HYB3 LIGHT CHAIN|HOMO SAPIENS (9606)
Pattern 1 | Position: 202 | STVEKT
=====================================================
1WF5_1|Chain A|sidekick 2 protein|Homo sapiens (9606)
Pattern 1 | Position: 29 | STVERR
=====================================================
1Z1D_1|Chain A|Replication protein A 32 kDa subunit|Homo sapiens (9606)
Pattern 1 | Position: 90 | STVDDD
=====================================================
1Z7Q_15|Chains c,d,e,f,g,h,i,j,k,l,m,n,o,p|proteasome activator protein PA26|
Trypanosoma brucei (5691)
Pattern 1 | Position: 153 | STVEDK
=====================================================
1ZTM_1|Chains A,B,C|Fusion glycoprotein|Human parainfluenza virus 3 (11216)
Pattern 1 | Position: 228 | STVDKY
=====================================================
1ZVO_1|Chains A,B|myeloma immunoglobulin D lambda|Homo sapiens (9606)
Pattern 1 | Position: 202 | STVEKT
=====================================================
2APC_1|Chain A|Alpha-1,3-mannosyl-glycoprotein 2-beta-N-
acetylglucosaminyltransferase|Oryctolagus cuniculus (9986)
Pattern 1 | Position: 13 | STVRRC
=====================================================
2B0S_1|Chain L|Fab 2219, light chain|Homo sapiens (9606)
Pattern 1 | Position: 206 | STVEKT
=====================================================
2B1A_1|Chain L|Fab 2219, light chain|Homo sapiens (9606)
Pattern 1 | Position: 206 | STVEKT
=====================================================
2B1H_1|Chain L|Fab 2219, light chain|Homo sapiens (9606)
Pattern 1 | Position: 206 | STVEKT
=====================================================
2BB0_1|Chains A,B|Imidazolonepropionase|Bacillus subtilis (1423)
```

```
Pattern 1 | Position: 114 | STVKDT
====================================================
2BB5_1|Chains A,B|Transcobalamin II|Homo sapiens (9606)
Pattern 1 | Position: 335 | STVEDV
====================================================
2DD8_2|Chain L|IGG Light Chain|Homo sapiens (9606)
Pattern 1 | Position: 201 | STVEKT
====================================================
2DVV_1|Chain A|Bromodomain-containing protein 2|Homo sapiens (9606)
Pattern 1 | Position: 54 | STVKRK
====================================================
2E3K_1|Chains A,B,C,D|Bromodomain-containing protein 2|Homo sapiens (9606)
Pattern 1 | Position: 54 | STVKRK
====================================================
2E7N_1|Chain A|Bromodomain-containing protein 3|Homo sapiens (9606)
Pattern 1 | Position: 61 | STVKRK
====================================================
2ES7_1|Chains A,B,C,D|putative thiol-disulfide isomerase and thioredoxin|Salmonella
typhimurium (99287)
Pattern 1 | Position: 25 | STVDDW
====================================================
2FB4_1|Chain L|IGG1-LAMBDA KOL FAB (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
====================================================
2FH6_1|Chain A|pullulanase|Klebsiella aerogenes (548)
Pattern 1 | Position: 522 | STVEEV
====================================================
2FH8_1|Chain A|pullulanase|Klebsiella aerogenes (548)
Pattern 1 | Position: 522 | STVEEV
====================================================
2FHB_1|Chain A|pullulanase|Klebsiella aerogenes (548)
Pattern 1 | Position: 522 | STVEEV
====================================================
2FHC_1|Chain A|pullulanase|Klebsiella aerogenes (548)
Pattern 1 | Position: 522 | STVEEV
====================================================
2FHF_1|Chain A|pullulanase|Klebsiella aerogenes (548)
Pattern 1 | Position: 522 | STVEEV
====================================================
2FL5_1|Chains A,C,E,L|Immunoglobulin Igg1 Lambda Light Chain|Homo sapiens (9606)
Pattern 1 | Position: 200 | STVEKT
====================================================
2G3F_1|Chains A,B|Imidazolonepropionase|Bacillus subtilis (1423)
Pattern 1 | Position: 114 | STVKDT
====================================================
2G4A_1|Chain A|Bromodomain-containing protein 2|Homo sapiens (9606)
Pattern 1 | Position: 50 | STVKRK
====================================================
2G75_2|Chains B,D|IGG Light Chain|Homo sapiens (9606)
Pattern 1 | Position: 201 | STVEKT
====================================================
2GAN_1|Chains A,B|182aa long hypothetical protein|Pyrococcus horikoshii (70601)
Pattern 1 | Position: 11 | STVKDE
====================================================
2H32_2|Chain B|Immunoglobulin omega chain|Homo sapiens (9606)
```

```
Pattern 1 | Position: 109 | STVEKT
====================================================
2H3N_2|Chains B,D|Ig lambda-5|Homo sapiens (9606)
Pattern 1 | Position: 108 | STVEKT
====================================================
2IDR_1|Chains A,B|Eukaryotic translation initiation factor 4E-1|
Triticum aestivum (4565)
Pattern 1 | Position: 35 | STVEDF
====================================================
2IDV_1|Chain A|Eukaryotic translation initiation factor 4E-1|Triticum aestivum (4565)
Pattern 1 | Position: 35 | STVEDF
====================================================
2IG2_1|Chain L|IGG1-LAMBDA KOL FAB (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
====================================================
2J28_23|Chain O|50S RIBOSOMAL PROTEIN L18|ESCHERICHIA COLI (562)
Pattern 1 | Position: 52 | STVEKA
====================================================
2J42_1|Chain A|C2 TOXIN COMPONENT-II|CLOSTRIDIUM BOTULINUM (1491)
Pattern 1 | Position: 338 | STVDDT
====================================================
2J6E_3|Chains L,M|IGM|HOMO SAPIENS (9606)
Pattern 1 | Position: 223 | STVEKT
====================================================
2JB5_2|Chain L|FAB FRAGMENT MOR03268 LIGHT CHAIN|HOMO SAPIENS (9606)
Pattern 1 | Position: 205 | STVEKT
====================================================
2JB6_1|Chains A,L|FAB FRAGMENT MOR03268 LIGHT CHAIN|HOMO SAPIENS (9606)
Pattern 1 | Position: 205 | STVEKT
====================================================
2JE8_1|Chains A,B|BETA-MANNOSIDASE|BACTEROIDES THETAIOTAOMICRON (226186)
Pattern 1 | Position: 349 | VTTERY
Pattern 1 | Position: 475 | STVKEF
====================================================
2JQ3_1|Chain A|Apolipoprotein C-III|Homo sapiens (9606)
Pattern 1 | Position: 55 | STVKDK
====================================================
2KC8_1|Chain A|Toxin relE|Escherichia coli (83333)
Pattern 1 | Position: 23 | STVREQ
====================================================
2KC9_1|Chain A|Toxin relE|Escherichia coli (83333)
Pattern 1 | Position: 23 | STVREQ
====================================================
2LDX_1|Chains A,B,C,D|APO-LACTATE DEHYDROGENASE|Mus musculus (10090)
Pattern 1 | Position: 1 | STVKEQ
====================================================
2MCG_1|Chains 1,2|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
====================================================
2MXC_1|Chain A|Sorting nexin-3|Homo sapiens (9606)
Pattern 1 | Position: 75 | STVRRR
====================================================
2OAJ_1|Chain A|Protein SNI1|Saccharomyces cerevisiae (4932)
Pattern 1 | Position: 753 | STVEKN
Pattern 1 | Position: 825 | STVKEQ
```

```
=====================================================
2OLD_1|Chains A,B|Bence Jones KWR Protein - Immunoglobulin Light Chain|Homo
sapiens (9606)
Pattern 1 | Position: 205 | STVEKT
=====================================================
2OMB_1|Chains A,B,C,D|Bence Jones KWR Protein - Immunoglobulin Light Chain|Homo
sapiens (9606)
Pattern 1 | Position: 205 | STVEKT
=====================================================
2OMN_1|Chains A,B|Bence Jones KWR Protein - Immunoglobulin Light Chain|
Homo sapiens (9606)
Pattern 1 | Position: 205 | STVEKT
=====================================================
2PI2_1|Chains A,B,C,D|Replication protein A 32 kDa subunit|Homo sapiens (9606)
Pattern 1 | Position: 257 | STVDDD
=====================================================
2QA2_1|Chain A|Polyketide oxygenase CabE|Streptomyces (1883)
Pattern 1 | Position: 166 | STVRKA
=====================================================
2RCJ_2|Chains C,D,G,H,K,L,O,P,S,T|IgA1 heavy chain|Homo sapiens (9606)
Pattern 1 | Position: 202 | STVEKT
=====================================================
2RDO_14|Chain O|50S ribosomal protein L18|Escherichia coli (562)
Pattern 1 | Position: 52 | STVEKA
=====================================================
2RDO_34|Chain 8|Ribosome recycling factor|Escherichia coli (562)
Pattern 1 | Position: 57 | VTVEDS
=====================================================
2VJX_1|Chains A,B|BETA-MANNOSIDASE|BACTEROIDES THETAIOTAOMICRON (226186)
Pattern 1 | Position: 347 | VTTERY
Pattern 1 | Position: 473 | STVKEF
=====================================================
2VL4_1|Chains A,B|BETA-MANNOSIDASE|BACTEROIDES THETAIOTAOMICRON (226186)
Pattern 1 | Position: 347 | VTTERY
Pattern 1 | Position: 473 | STVKEF
=====================================================
2VMF_1|Chains A,B|BETA-MANNOSIDASE|BACTEROIDES THETAIOTAOMICRON (226186)
Pattern 1 | Position: 347 | VTTERY
Pattern 1 | Position: 473 | STVKEF
=====================================================
2VO5_1|Chains A,B|BETA-MANNOSIDASE|BACTEROIDES THETAIOTAOMICRON (226186)
Pattern 1 | Position: 347 | VTTERY
Pattern 1 | Position: 473 | STVKEF
=====================================================
2VOT_1|Chains A,B|BETA-MANNOSIDASE|BACTEROIDES THETAIOTAOMICRON (226186)
Pattern 1 | Position: 347 | VTTERY
Pattern 1 | Position: 473 | STVKEF
=====================================================
2VQT_1|Chains A,B|BETA-MANNOSIDASE|BACTEROIDES THETAIOTAOMICRON (818)
Pattern 1 | Position: 347 | VTTERY
Pattern 1 | Position: 473 | STVKEF
=====================================================
2VQU_1|Chains A,B|BETA-MANNOSIDASE|BACTEROIDES THETAIOTAOMICRON (818)
Pattern 1 | Position: 347 | VTTERY
Pattern 1 | Position: 473 | STVKEF
```

```
==================================================
2VR4_1|Chains A,B|BETA-MANNOSIDASE|BACTEROIDES THETAIOTAOMICRON (226186)
Pattern 1 | Position: 347 | VTTERY
Pattern 1 | Position: 473 | STVKEF
==================================================
1AAO_1|Chain A|FIBRITIN|Enterobacteria phage T4 (10665)
Pattern 1 | Position: 41 | STVEER
==================================================
1FO8_1|Chain A|ALPHA-1,3-MANNOSYL-GLYCOPROTEIN BETA-1,2-N-
ACETYLGLUCOSAMINYLTRANSFERASE|Oryctolagus cuniculus (9986)
Pattern 1 | Position: 14 | STVRRC
==================================================
1JKM_1|Chains A,B|BREFELDIN A ESTERASE|Bacillus subtilis (1423)
Pattern 1 | Position: 345 | STVRDV
==================================================
1JWO_2|Chain B|cephalosporin acylase beta chain|Brevundimonas diminuta (293)
Pattern 1 | Position: 112 | STVDKP
==================================================
1MCD_1|Chains A,B|Immunoglobulin lambda-1 light chain|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCI_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCK_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
1MCN_1|Chains A,B|IMMUNOGLOBULIN LAMBDA DIMER MCG (LIGHT CHAIN)|Homo sapiens (9606)
Pattern 1 | Position: 204 | STVEKT
==================================================
```

Question 6

Selecting all transmembrane proteins, with "beta-barrel" in their name (as shown in the query). We get 13,090 results.

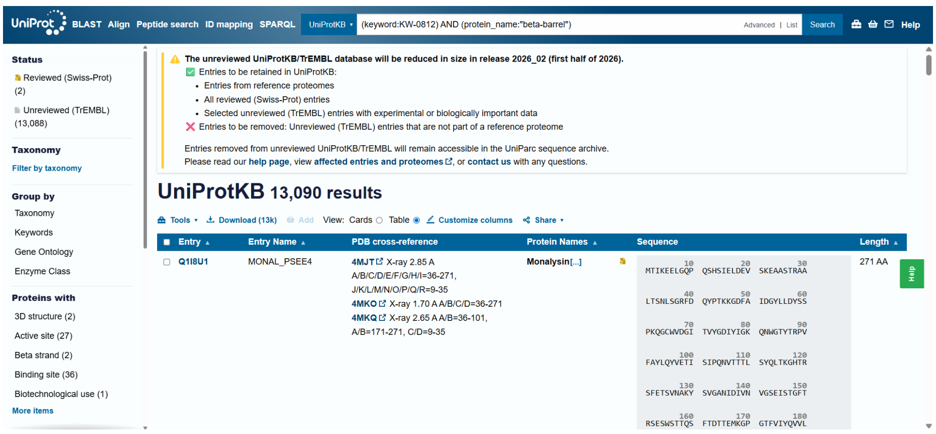

Figure 6.1: Sequence data of beta-barrel membrane proteins

Sequence Manipulation Suite

The pattern **[KRHQFE].G[IVLFAC].[IVLFMYW].[IVLFW]** was used to find the beta-barrel proteins with the required sequence and the result is shown below:

Figure 6.2: Sequence Manipulation Suite obtained 5043 matching sequences

We infer the number of matching sequences as $13,090 - 8,047 = 5,043$ sequences.

ScanProsite

Since the database contains more than 1,000 sequences, we have to upload the FASTA file to ScanProsite so that it can be analyzed, the process is shown below:

Figure 6.3: Submitting the file to the ScanProsite Server

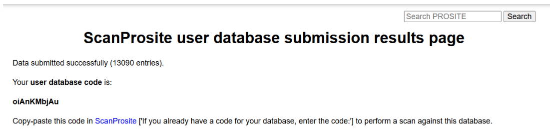

Figure 6.4: ScanProsite Database Code

We use the uploaded database to find the sequences with our pattern subsequence using the pattern **[K,R,H,Q,F,E]-x-G-[I,V,L,F,A,C]-x-[I,V,L,F,M,Y,W]-x-[I,V,L,F,W]** as shown below:

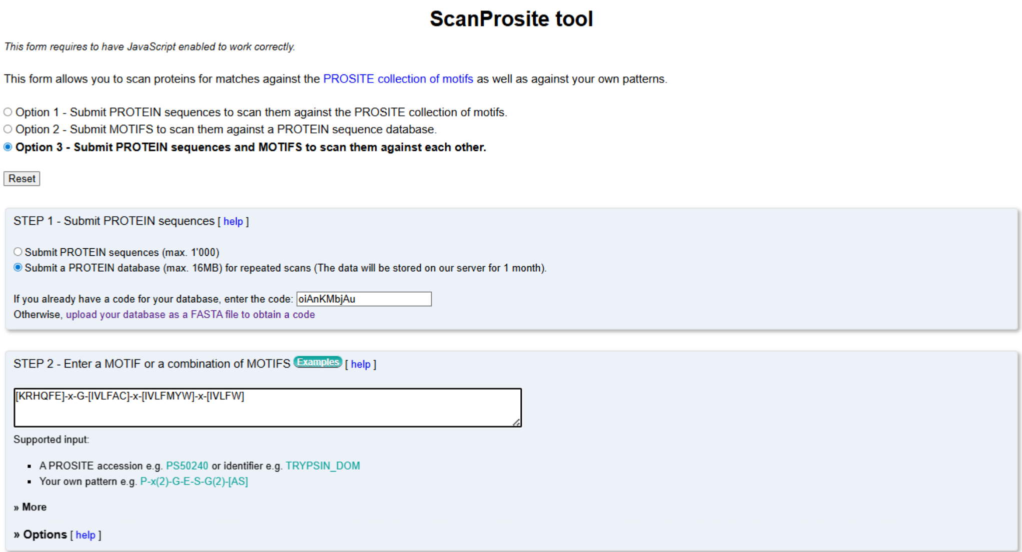

Figure 6.5: ScanProsite Command to search for the required pattern

The results obtained from ScanProsite are shown below:

Gmail

Varunkumar Krishnakumar <varunkumar0135@gmail.com>

[ScanProsite] Results for job: Pattern_Assignment-8_Question-6

Expasy <expasy@expasy.org>
To: varunkumar0135@gmail.com

Tue, Mar 24, 2026 at 1:57 AM

[Mon Mar 23 17:57:17 2026] - Submission
[Mon Mar 23 21:27:45 2026] - Results:

Output format: Text

Hits for USER001 "[KRHQFE]-x-G-[IVLFAC]-x-[IVLFMYW]-x-[IVLFW]" on oiAnKMbjAu custom database sequences

Found: 7464 hits in 5042 sequences<hr>

[View matches in UniProtKB]
<hr>
Graphical View (graphical view with feature detection): https://prosite.expasy.org/cgi-bin/prosite/ScanView.cgi?scanfile=2495254.scan.gz&sig=[KRHQFE]-x-G-[IVLFAC]-x-[IVLFMYW]-x-[IVLFW]
(link will be valid for 12hours)

>USER001 (user pattern):
Pattern: [KRHQFE]-x-G-[IVLFAC]-x-[IVLFMYW]-x-[IVLFW]
Approximate number of expected random matches in ~ 100'000 sequences (50'000'000 residues): 26784

>tr|A0A009I6C7|A0A009I6C7_ACIB9
376 - 383: QaGIkYkl

>tr|A0A014NQX0|A0A014NQX0_9GAMM (524 aa)
TonB-dependent receptor-like beta-barrel domain-containing protein. [Erwinia mallotivora]
MEVQLGNFNRRQVQFDISDKLDEDGEWLYRMVGLARDSDTQFQYRDGPDVEDNRVYLAPTVTWIPSADTSLTLRADYLRD
TSGGTIAVLTTPGGKVTDTLLGDYSYNRFRQEQYSVGYEFSQQLAESVKFSQNLRYGQTDTILNNLLPGDPDFTTDSVSR
NAIRFDEHMKAFNVDNQLQMDFNTATVAHTLLAGVDYSWLDANAKRFSAVAPALSLSNPLYGLDISDPTNAVNNNDQTTQ
QLGVYLQDQIHLTEHWLLTLGGRHDDISMRTQNNLTGNTSWIDKDAWTGRAGLTWLVGNGLAPYISYAESFVPNSGTDSQ
NRTFDPSKAHQYEMGVKYQPDSNLLMTLAAFEITKSNVLTNEIVNGSATGYQVASGEVRSRGIEWEAQAKLTDNLDALAS
YSLTDAQITKSNNGDEGNQQANVPKNMASVWVNYGFHQGLLDGLMLSSGVRYIGSMYGDNANTIQVKNVTLMDAGVKYQV
NKNIQVGFNVQNLFDRHYVGTCDGDTSCYPGQERTFIGSLKLNF
290 - 297: RaGLtWIV

Figure 6.5: ScanProsite Results

ScanProsite found: 7464 hits in 5042 sequences.

Link to the google drive containing the code and the result files from the different tools used in this assignment: `https://drive.google.com/drive/u/0/folders/1Tp2v_ITol8HNKb_I5gwO52r5IMriJtQ5`

Annotations have been omitted from the submitted PDF for the sake of brevity. The fully annotated version of the code is available in the code folder within the attached Google Drive.
